

Big Data Project

Hadoop HDFS Commands Practice

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Dataset Name:	Synthetic E-Commerce Transactions Dataset
Dataset File:	dataset.csv
Dataset Size:	Greater than 500 MB
Environment:	Instructor Hadoop Virtual Machine / VirtualBox
Submission Date:	8 May 2026

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Project Description

This project practices Hadoop Distributed File System (HDFS) commands using a large, structured CSV dataset. The dataset is uploaded to HDFS, inspected with basic commands, managed with file operations, and explored using additional HDFS utilities. The objective is to gain hands-on experience with the Hadoop ecosystem and understand how to interact with HDFS through the command-line interface.

Dataset Description

The dataset is a synthetic e-commerce transactions file named `dataset.csv`. Each row represents an order and includes fields such as order date, customer segment, country, city, product category, quantity, unit price, discount, shipping mode, status, rating, total amount, profit, and marketing channel. It is structured CSV data suitable for HDFS upload and command-line exploration.

Dataset source: Generated locally using the `generate_ecommerce_dataset.py` Python script. The dataset contains 2,950,001 lines (including header) and occupies 531.6 MB in HDFS, making it an excellent candidate for big data processing practice.

Environment Setup

The project uses the instructor-provided Hadoop virtual machine running in VirtualBox. After starting the VM, Hadoop services are verified using the `jps` command. The expected services include NameNode and DataNode for HDFS, as well as ResourceManager and NodeManager for YARN.

Starting Hadoop Services

The following screenshot shows the Hadoop services being started using `start-dfs.sh` and `start-yarn.sh` commands. The `jps` output confirms all required Java processes are running: NameNode (PID 3312), ResourceManager (PID 3908), SecondaryNameNode (PID 3657), NodeManager (PID 4027), and DataNode (PID 3486).

```

2 mai 22:14
test@test-VirtualBox: ~
-rwxrwx--- 1 root vboxsf 532M mai 2 22:02 dataset.csv
-rwxrwx--- 1 root vboxsf 4,5K mai 2 22:02 generate_ecommerce_dataset.py
-rwxrwx--- 1 root vboxsf 94M mai 2 22:03 Hadoop_HDFS_Project_Submission.zip
-rwxrwx--- 1 root vboxsf 3,8K mai 2 22:02 Hadoop_HDFS_VM_Guide.md
-rwxrwx--- 1 root vboxsf 8,5K mai 2 22:03 Report.pdf
-rwxrwx--- 1 root vboxsf 2,5K mai 2 22:02 run_hdfs_project.sh
drwxrwx--- 1 root vboxsf 0 mai 2 22:09 Screenshots
/usr/local/hadoop/bin/hdfs
3096 Jps
test@test-VirtualBox:~$ start-dfs.sh; start-yarn.sh; jps
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [test-VirtualBox]
2026-05-02 22:13:37,745 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers
3312 NameNode
3908 ResourceManager
4164 Jps
3657 SecondaryNameNode
4027 NodeManager
3486 DataNode
test@test-VirtualBox:~$

```

Figure 1: Hadoop services running - jps output showing all required processes

Shared Folder and Project Files

The shared folder `/media/sf_HadoopProject` contains all project files including the dataset (dataset.csv, 532 MB), the dataset generation script, the project submission zip, the VM guide, and the automation script (run_hdfs_project.sh).

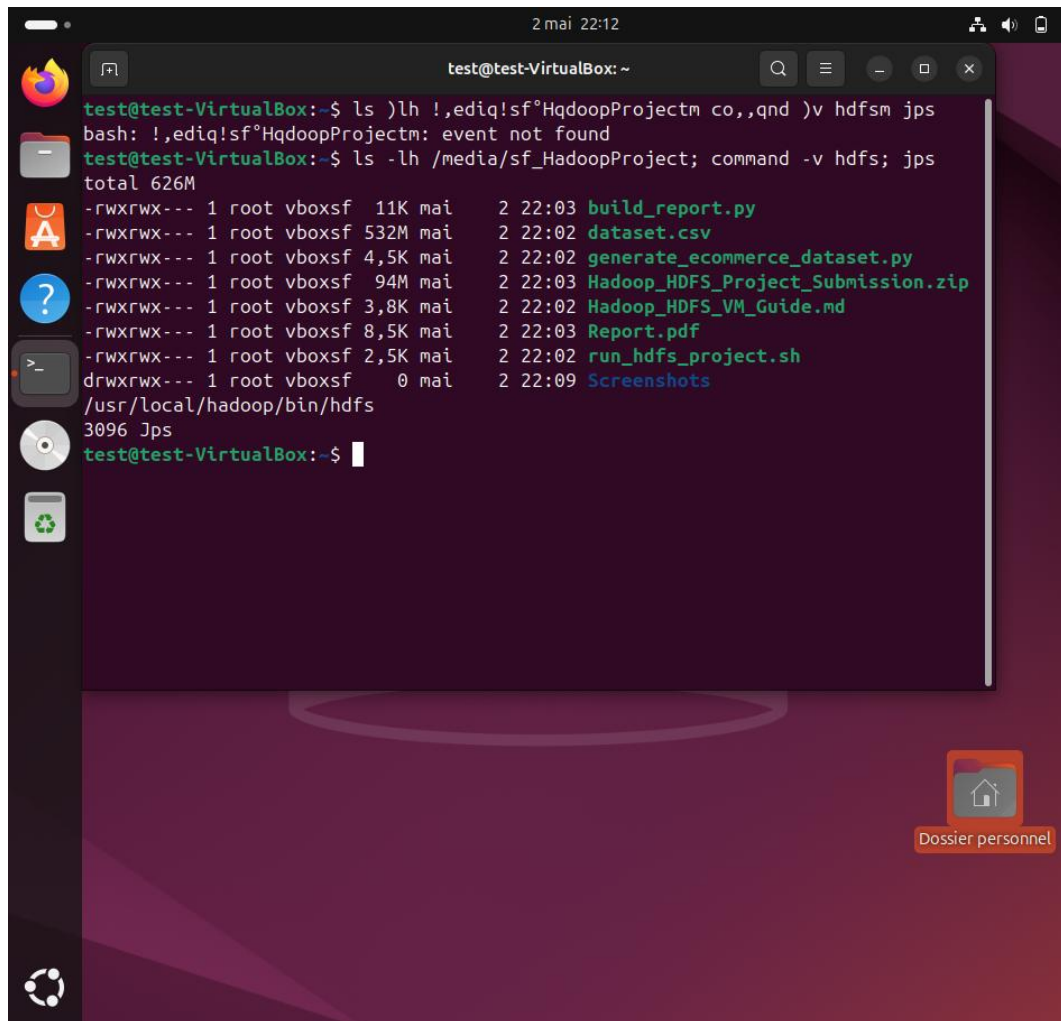


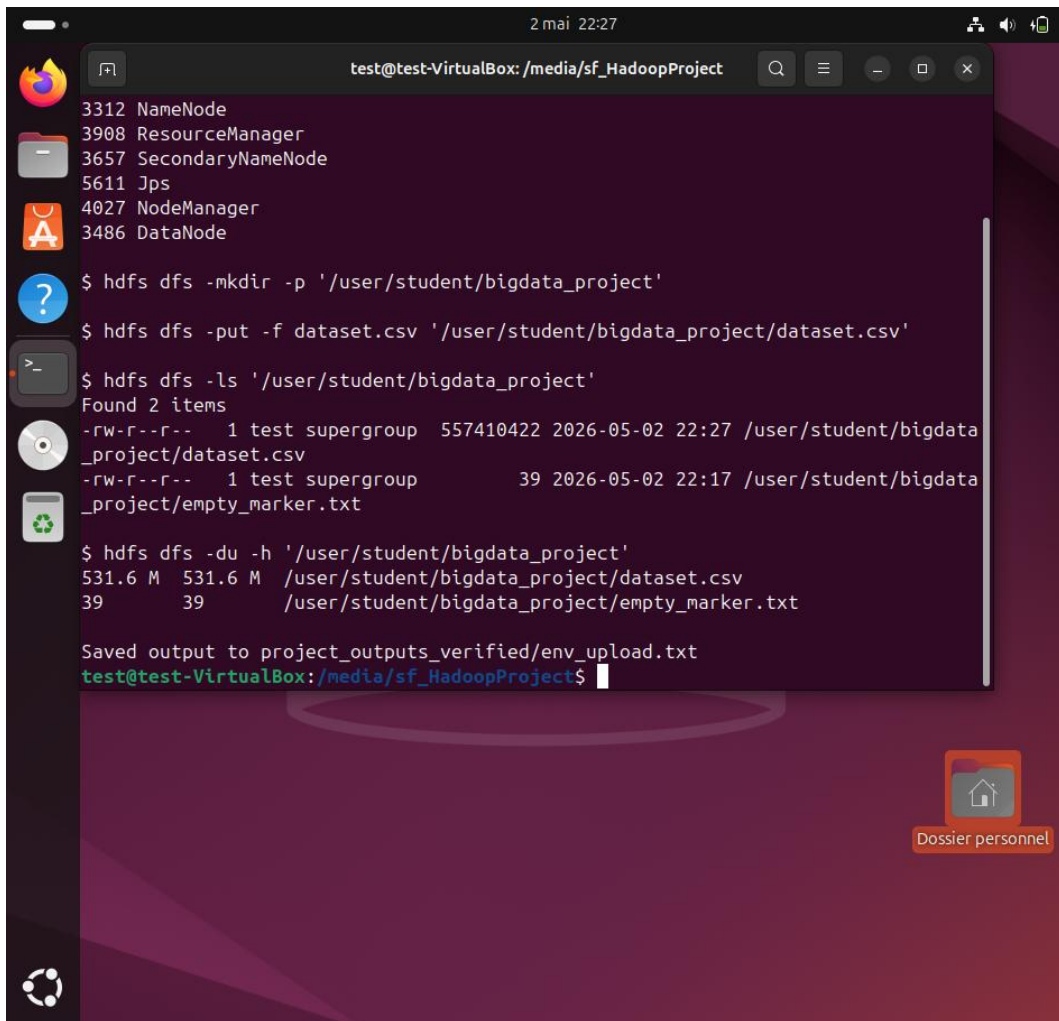
Figure 2: Shared folder contents showing project files and dataset

Upload Dataset to HDFS

The dataset is uploaded to HDFS using a series of commands. First, the project directory structure is created in HDFS, then the CSV file is transferred from the local filesystem to HDFS.

Command	Purpose / Expected Result
<code>hdfs dfs -mkdir -p /user/student/bigdata_project</code>	Creates the HDFS project directory recursively
<code>hdfs dfs -put -f dataset.csv /user/student/bigdata_project/dataset.csv</code>	Uploads the CSV dataset from local storage to HDFS (force overwrite)
<code>hdfs dfs -ls /user/student/bigdata_project</code>	Verifies that the dataset exists in HDFS
<code>hdfs dfs -du -h /user/student/bigdata_project</code>	Shows human-readable HDFS disk usage for the project directory

The screenshot below demonstrates the successful execution of environment setup and dataset upload commands. The output confirms that dataset.csv (531.6 MB) has been successfully uploaded to HDFS, along with an empty marker file.

A screenshot of a terminal window titled 'test@test-VirtualBox: /media/sf_HadoopProject'. The terminal shows the following commands and output:

```
3312 NameNode
3908 ResourceManager
3657 SecondaryNameNode
5611 Jps
4027 NodeManager
3486 DataNode

$ hdfs dfs -mkdir -p '/user/student/bigdata_project'

$ hdfs dfs -put -f dataset.csv '/user/student/bigdata_project/dataset.csv'

$ hdfs dfs -ls '/user/student/bigdata_project'
Found 2 items
-rw-r--r--  1 test supergroup  557410422 2026-05-02 22:27 /user/student/bigdata_project/dataset.csv
-rw-r--r--  1 test supergroup      39 2026-05-02 22:17 /user/student/bigdata_project/empty_marker.txt

$ hdfs dfs -du -h '/user/student/bigdata_project'
531.6 M  531.6 M  /user/student/bigdata_project/dataset.csv
39      39      /user/student/bigdata_project/empty_marker.txt

Saved output to project_outputs_verified/env_upload.txt
test@test-VirtualBox: /media/sf_HadoopProject$
```

The terminal window is part of a desktop environment with a dark purple background. On the left, there is a sidebar with icons for Firefox, a file manager, a terminal, a CD, and a trash. On the right, there is a 'Dossier personnel' icon. The top of the window shows the date and time as '2 mai 22:27'.

Figure 3: Environment setup and dataset upload to HDFS - showing successful upload of dataset.csv

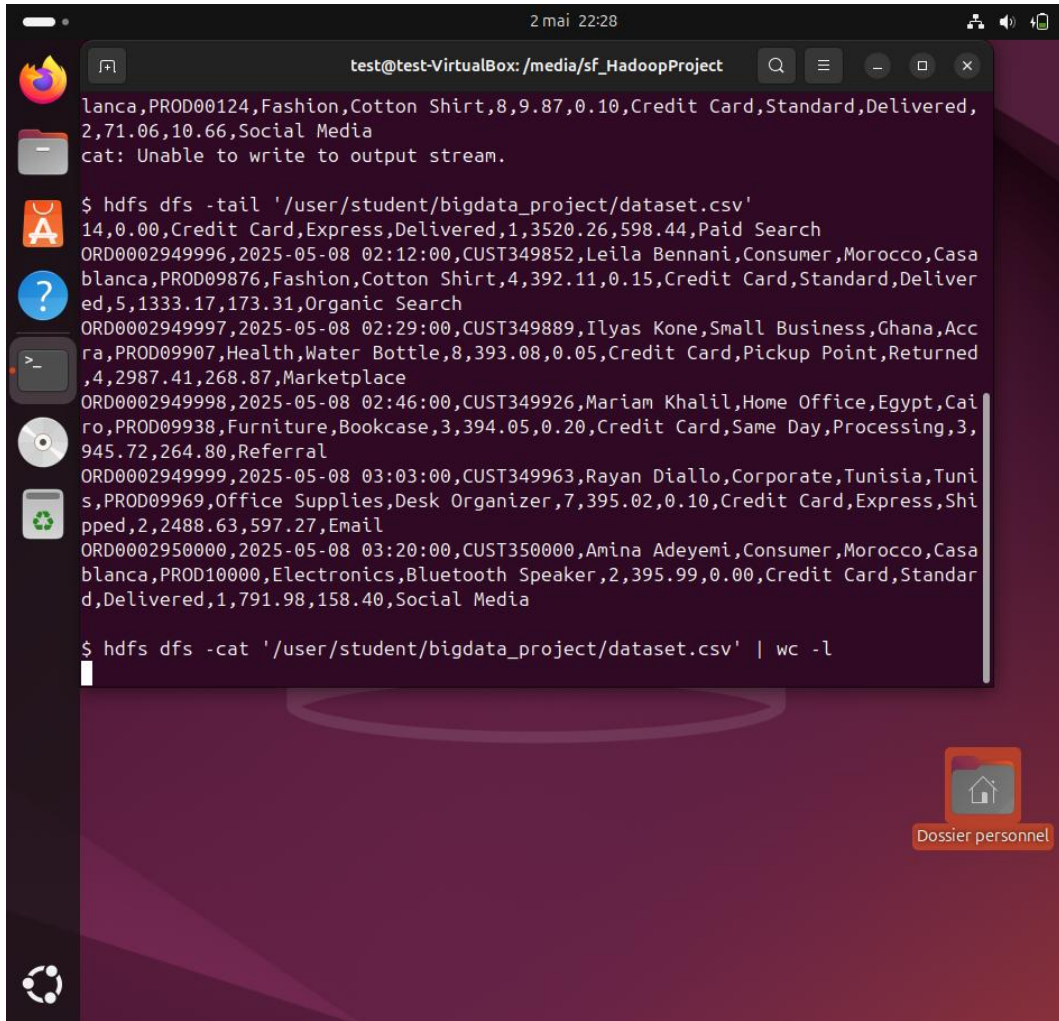
Basic HDFS Commands

This section demonstrates fundamental HDFS commands used to inspect and explore the uploaded dataset. These commands are essential for day-to-day interaction with HDFS.

Command	Purpose / Expected Result
<code>hdfs dfs -ls /</code>	Lists files and directories at the HDFS root
<code>hdfs dfs -cat /user/student/bigdata_project/dataset.csv head -n 5</code>	Displays the CSV header and first records without printing the whole file
<code>hdfs dfs -tail /user/student/bigdata_project/dataset.csv</code>	Displays the last part of the dataset
<code>hdfs dfs -cat /user/student/bigdata_project/dataset.csv wc -l</code>	Counts the total number of lines in the dataset
<code>hdfs dfs -du -h /user/student/bigdata_project</code>	Shows human-readable HDFS disk usage
<code>hdfs dfs -count /user/student/bigdata_project</code>	Shows directory, file, and byte counts

Content Inspection Commands

The following screenshot shows the execution of basic content commands including `cat` piped to `head` for previewing the dataset structure, `tail` for viewing the end of the file, and the line count command in progress.



```

test@test-VirtualBox: /media/sf_HadoopProject
lanca,PROD00124,Fashion,Cotton Shirt,8,9.87,0.10,Credit Card,Standard,Delivered,
2,71.06,10.66,Social Media
cat: Unable to write to output stream.

$ hdfs dfs -tail '/user/student/bigdata_project/dataset.csv'
14,0.00,Credit Card,Express,Delivered,1,3520.26,598.44,Paid Search
ORD0002949996,2025-05-08 02:12:00,CUST349852,Leila Bennani,Consumer,Morocco,Casa
blanca,PROD09876,Fashion,Cotton Shirt,4,392.11,0.15,Credit Card,Standard,Deliver
ed,5,1333.17,173.31,Organic Search
ORD0002949997,2025-05-08 02:29:00,CUST349889,Ilyas Kone,Small Business,Ghana,Acc
ra,PROD09907,Health,Water Bottle,8,393.08,0.05,Credit Card,Pickup Point,Returned
,4,2987.41,268.87,Marketplace
ORD0002949998,2025-05-08 02:46:00,CUST349926,Mariam Khalil,Home Office,Egypt,Cai
ro,PROD09938,Furniture,Bookcase,3,394.05,0.20,Credit Card,Same Day,Processing,3,
945.72,264.80,Referral
ORD0002949999,2025-05-08 03:03:00,CUST349963,Rayan Diallo,Corporate,Tunisia,Tuni
s,PROD09969,Office Supplies,Desk Organizer,7,395.02,0.10,Credit Card,Express,Shi
pped,2,2488.63,597.27,Email
ORD0002950000,2025-05-08 03:20:00,CUST350000,Amina Adeyemi,Consumer,Morocco,Casa
blanca,PROD10000,Electronics,Bluetooth Speaker,2,395.99,0.00,Credit Card,Standar
d,Delivered,1,791.98,158.40,Social Media

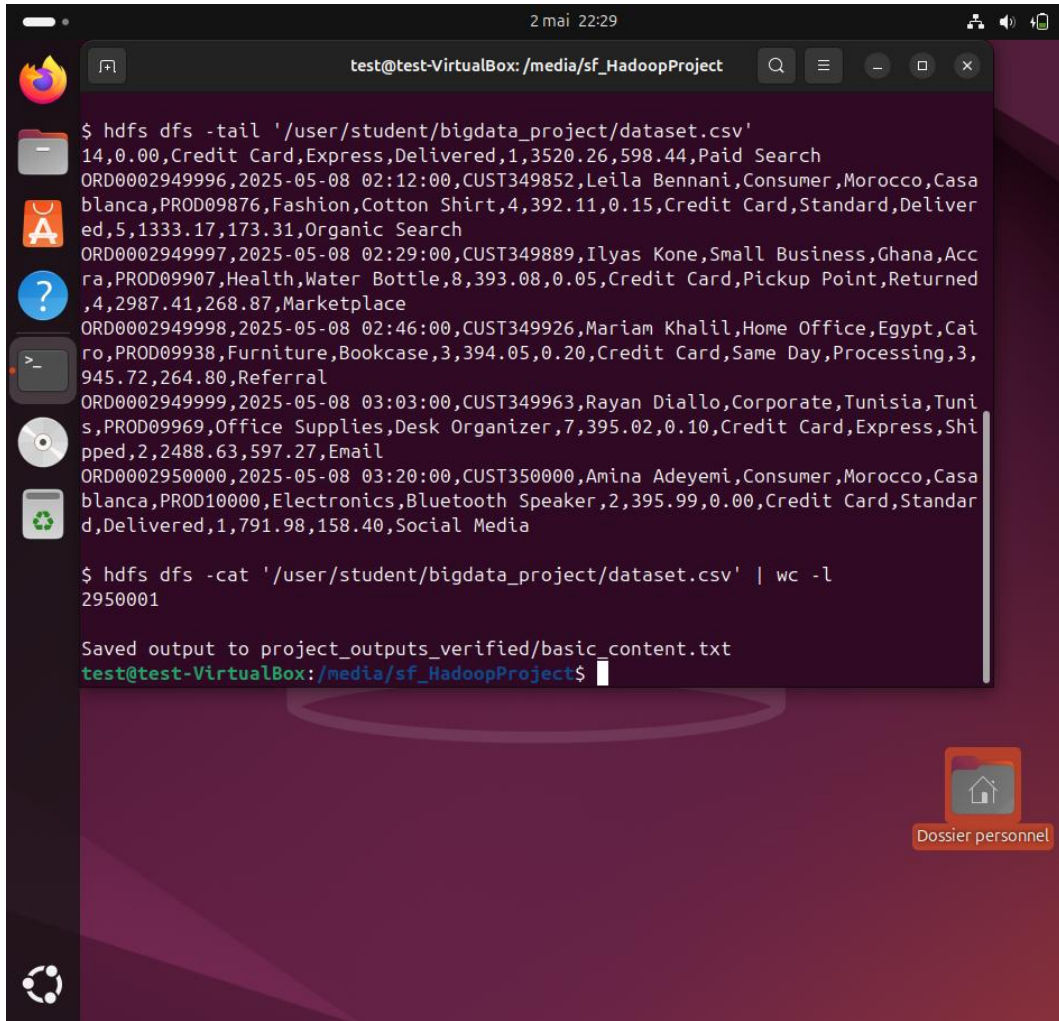
$ hdfs dfs -cat '/user/student/bigdata_project/dataset.csv' | wc -l

```

Figure 4: Basic content commands - cat/head preview and tail showing the last records of the dataset

Line Count Result

The line count command confirms that the dataset contains exactly 2,950,001 lines, including the CSV header row. This large dataset size validates the project's big data requirements.



```

test@test-VirtualBox: /media/sf_HadoopProject
$ hdfs dfs -tail '/user/student/bigdata_project/dataset.csv'
14,0.00,Credit Card,Express,Delivered,1,3520.26,598.44,Paid Search
ORD0002949996,2025-05-08 02:12:00,CUST349852,Leila Bennani,Consumer,Morocco,Casa
blanca,PROD09876,Fashion,Cotton Shirt,4,392.11,0.15,Credit Card,Standard,Deliver
ed,5,1333.17,173.31,Organic Search
ORD0002949997,2025-05-08 02:29:00,CUST349889,Ilyas Kone,Small Business,Ghana,Acc
ra,PROD09907,Health,Water Bottle,8,393.08,0.05,Credit Card,Pickup Point,Returned
,4,2987.41,268.87,Marketplace
ORD0002949998,2025-05-08 02:46:00,CUST349926,Mariam Khalil,Home Office,Egypt,Cai
ro,PROD09938,Furniture,Bookcase,3,394.05,0.20,Credit Card,Same Day,Processing,3,
945.72,264.80,Referral
ORD0002949999,2025-05-08 03:03:00,CUST349963,Rayan Diallo,Corporate,Tunisia,Tuni
s,PROD09969,Office Supplies,Desk Organizer,7,395.02,0.10,Credit Card,Express,Shi
pped,2,2488.63,597.27,Email
ORD0002950000,2025-05-08 03:20:00,CUST350000,Amina Adeyemi,Consumer,Morocco,Casa
blanca,PROD10000,Electronics,Bluetooth Speaker,2,395.99,0.00,Credit Card,Standar
d,Delivered,1,791.98,158.40,Social Media

$ hdfs dfs -cat '/user/student/bigdata_project/dataset.csv' | wc -l
2950001

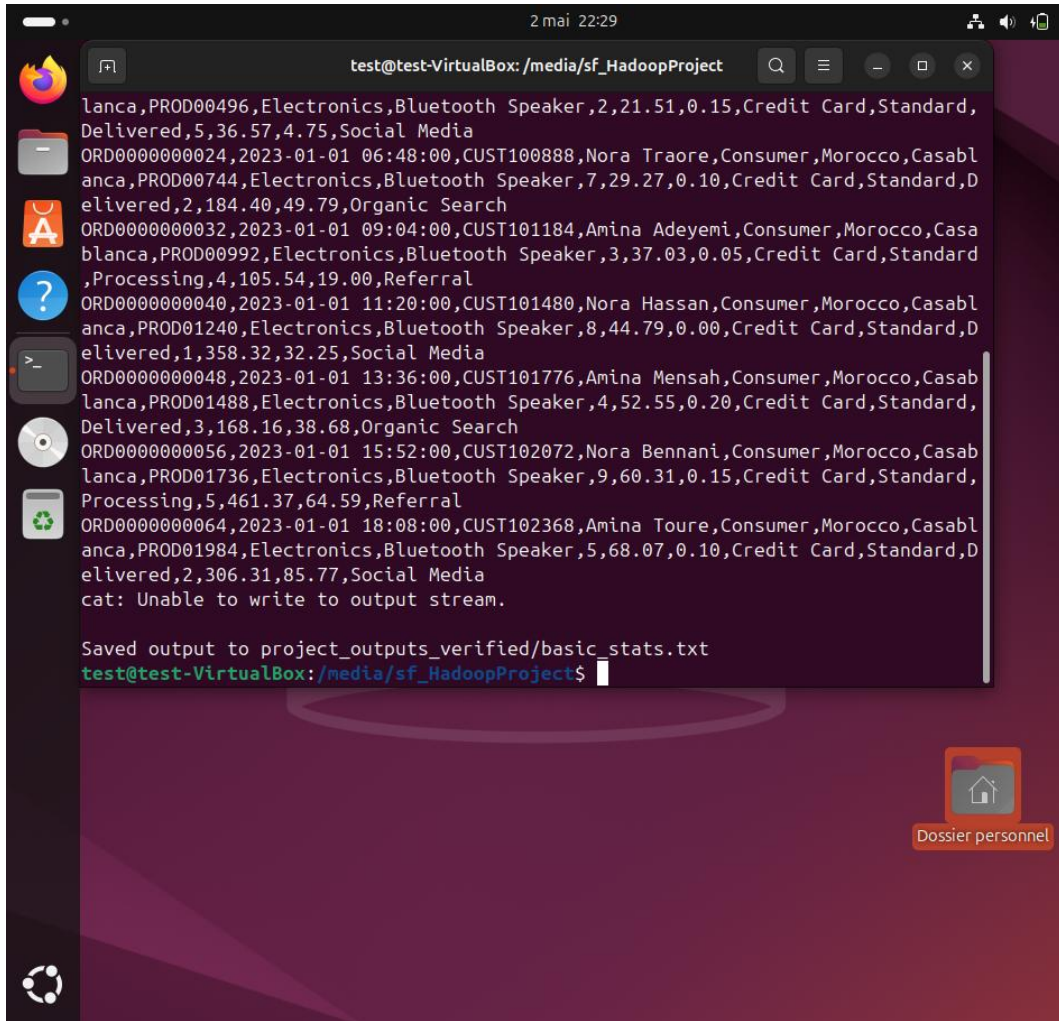
Saved output to project_outputs_verified/basic_content.txt
test@test-VirtualBox: /media/sf_HadoopProject$

```

Figure 5: Line count result - dataset contains 2,950,001 lines including header

Grep Filter for Electronics Category

The grep command filters rows containing the keyword 'Electronics' from the dataset. This demonstrates how to perform basic data filtering directly on HDFS without downloading the file. The output shows multiple electronics product records including Bluetooth Speaker products sold in Morocco.



```

test@test-VirtualBox: /media/sf_HadoopProject
lanca,PROD00496,Electronics,Bluetooth Speaker,2,21.51,0.15,Credit Card,Standard,
Delivered,5,36.57,4.75,Social Media
ORD0000000024,2023-01-01 06:48:00,CUST100888,Nora Traore,Consumer,Morocco,Casabl
anca,PROD00744,Electronics,Bluetooth Speaker,7,29.27,0.10,Credit Card,Standard,D
elivered,2,184.40,49.79,Organic Search
ORD0000000032,2023-01-01 09:04:00,CUST101184,Amina Adeyemi,Consumer,Morocco,Casa
blanca,PROD00992,Electronics,Bluetooth Speaker,3,37.03,0.05,Credit Card,Standard
,Processing,4,105.54,19.00,Referral
ORD0000000040,2023-01-01 11:20:00,CUST101480,Nora Hassan,Consumer,Morocco,Casabl
anca,PROD01240,Electronics,Bluetooth Speaker,8,44.79,0.00,Credit Card,Standard,D
elivered,1,358.32,32.25,Social Media
ORD0000000048,2023-01-01 13:36:00,CUST101776,Amina Mensah,Consumer,Morocco,Casab
lanca,PROD01488,Electronics,Bluetooth Speaker,4,52.55,0.20,Credit Card,Standard,
Delivered,3,168.16,38.68,Organic Search
ORD0000000056,2023-01-01 15:52:00,CUST102072,Nora Bennani,Consumer,Morocco,Casab
lanca,PROD01736,Electronics,Bluetooth Speaker,9,60.31,0.15,Credit Card,Standard,
Processing,5,461.37,64.59,Referral
ORD0000000064,2023-01-01 18:08:00,CUST102368,Amina Toure,Consumer,Morocco,Casabl
anca,PROD01984,Electronics,Bluetooth Speaker,5,68.07,0.10,Credit Card,Standard,D
elivered,2,306.31,85.77,Social Media
cat: Unable to write to output stream.

Saved output to project_outputs_verified/basic_stats.txt
test@test-VirtualBox: /media/sf_HadoopProject$

```

Figure 6: Grep filter output showing Electronics category records from the dataset

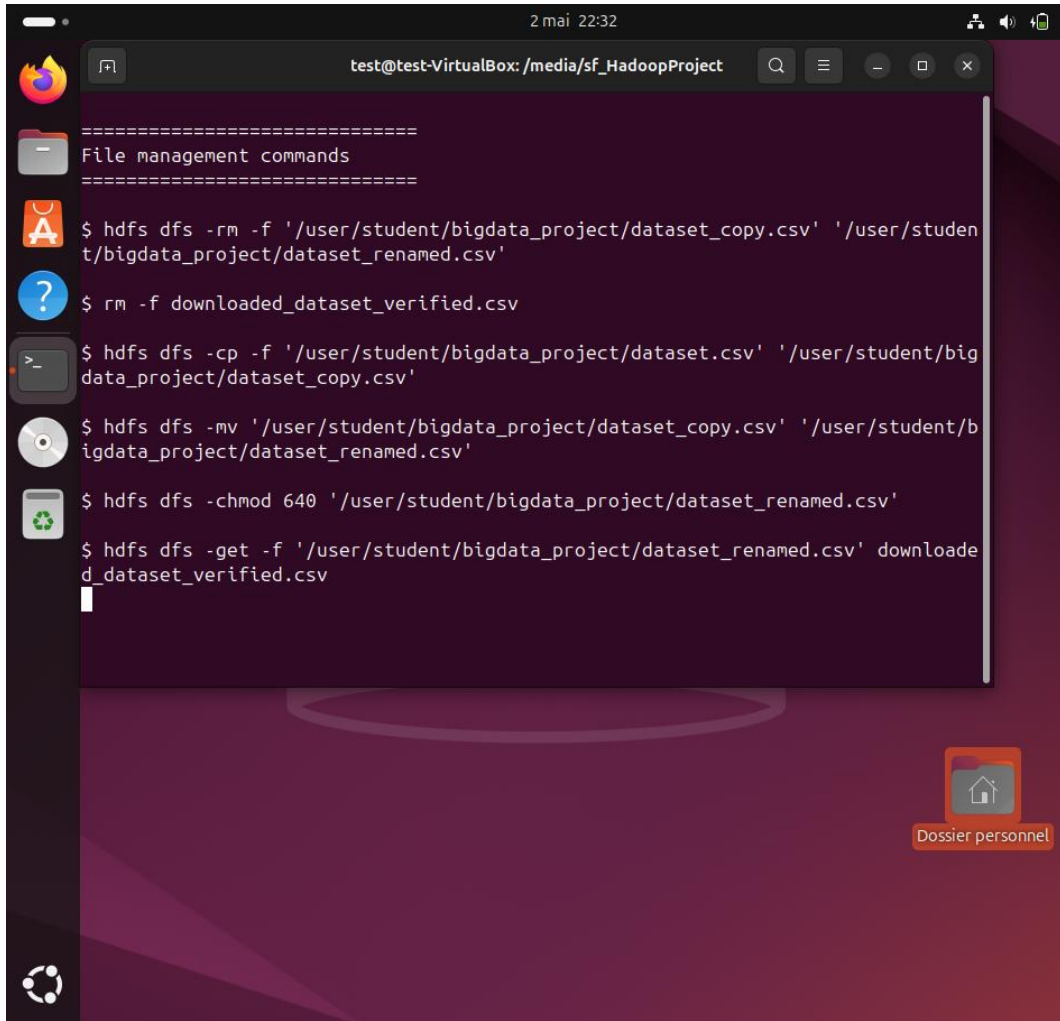
File Management Commands

This section demonstrates file management operations within HDFS, including copying, renaming, changing permissions, downloading, and deleting files.

Command	Purpose / Expected Result
<code>hdfs dfs -cp -f /user/student/bigdata_project/dataset.csv /user/student/bigdata_project/dataset_copy.csv</code>	Copies the dataset inside HDFS
<code>hdfs dfs -mv /user/student/bigdata_project/dataset_copy.csv /user/student/bigdata_project/dataset_renamed.csv</code>	Renames the copied HDFS file
<code>hdfs dfs -chmod 640 /user/student/bigdata_project/dataset_renamed.csv</code>	Changes permissions: owner read/write, group read
<code>hdfs dfs -get -f /user/student/bigdata_project/dataset_renamed.csv downloaded_dataset_verified.csv</code>	Downloads an HDFS file back to local storage
<code>hdfs dfs -rm /user/student/bigdata_project/dataset_renamed.csv</code>	Deletes the renamed copy from HDFS

File Management Execution

The first screenshot shows the execution of file management commands including cleanup of previous files, copy, move, chmod, and get operations.



The screenshot shows a terminal window titled 'test@test-VirtualBox: /media/sf_HadoopProject'. The terminal displays a series of HDFS commands for file management. On the left side of the terminal window, there is a vertical toolbar with icons for file management (a folder), application (a magnifying glass), help (a question mark), terminal (a prompt character), disk (a hard drive), and trash (a trash can). The commands entered in the terminal are as follows:

```

=====
File management commands
=====
$ hdfs dfs -rm -f '/user/student/bigdata_project/dataset_copy.csv' '/user/student/bigdata_project/dataset_renamed.csv'
$ rm -f downloaded_dataset_verified.csv
$ hdfs dfs -cp -f '/user/student/bigdata_project/dataset.csv' '/user/student/bigdata_project/dataset_copy.csv'
$ hdfs dfs -mv '/user/student/bigdata_project/dataset_copy.csv' '/user/student/bigdata_project/dataset_renamed.csv'
$ hdfs dfs -chmod 640 '/user/student/bigdata_project/dataset_renamed.csv'
$ hdfs dfs -get -f '/user/student/bigdata_project/dataset_renamed.csv' downloaded_dataset_verified.csv

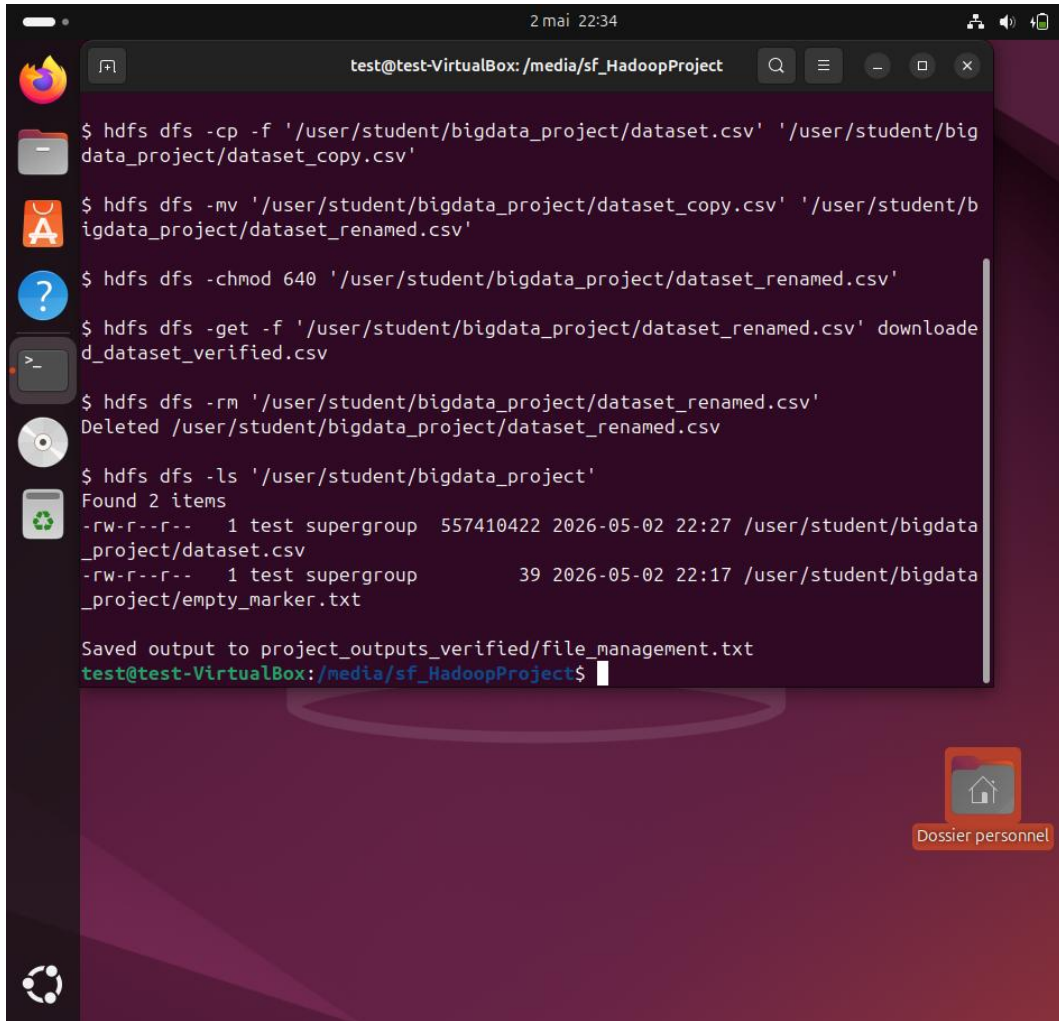
```

The terminal output shows the successful execution of these commands. The desktop background is a dark purple gradient with a 'Dossier personnel' icon in the bottom right corner.

Figure 7: File management commands - copy, rename, chmod, and get operations

File Management Completion

The second screenshot shows the completion of file management operations including the deletion of the renamed file and the final directory listing, which confirms only the original dataset.csv and empty_marker.txt remain.



The screenshot shows a terminal window titled 'test@test-VirtualBox: /media/sf_HadoopProject' with a dark background. The terminal displays a series of HDFS commands and their outputs. The commands performed are: copying a file, renaming it, changing permissions, downloading it, deleting it, and listing the directory. The final directory listing shows two files: 'dataset.csv' and 'empty_marker.txt'. The terminal also shows a message indicating the output was saved to a file. The desktop background is dark purple with a 'Dossier personnel' icon in the bottom right corner.

```
test@test-VirtualBox: /media/sf_HadoopProject
$ hdfs dfs -cp -f '/user/student/bigdata_project/dataset.csv' '/user/student/bigdata_project/dataset_copy.csv'
$ hdfs dfs -mv '/user/student/bigdata_project/dataset_copy.csv' '/user/student/bigdata_project/dataset_renamed.csv'
$ hdfs dfs -chmod 640 '/user/student/bigdata_project/dataset_renamed.csv'
$ hdfs dfs -get -f '/user/student/bigdata_project/dataset_renamed.csv' downloaded_dataset_verified.csv
$ hdfs dfs -rm '/user/student/bigdata_project/dataset_renamed.csv'
Deleted /user/student/bigdata_project/dataset_renamed.csv
$ hdfs dfs -ls '/user/student/bigdata_project'
Found 2 items
-rw-r--r--  1 test supergroup  557410422 2026-05-02 22:27 /user/student/bigdata_project/dataset.csv
-rw-r--r--  1 test supergroup           39 2026-05-02 22:17 /user/student/bigdata_project/empty_marker.txt

Saved output to project_outputs_verified/file_management.txt
test@test-VirtualBox: /media/sf_HadoopProject$
```

Figure 8: File management completed - rm command and final directory listing

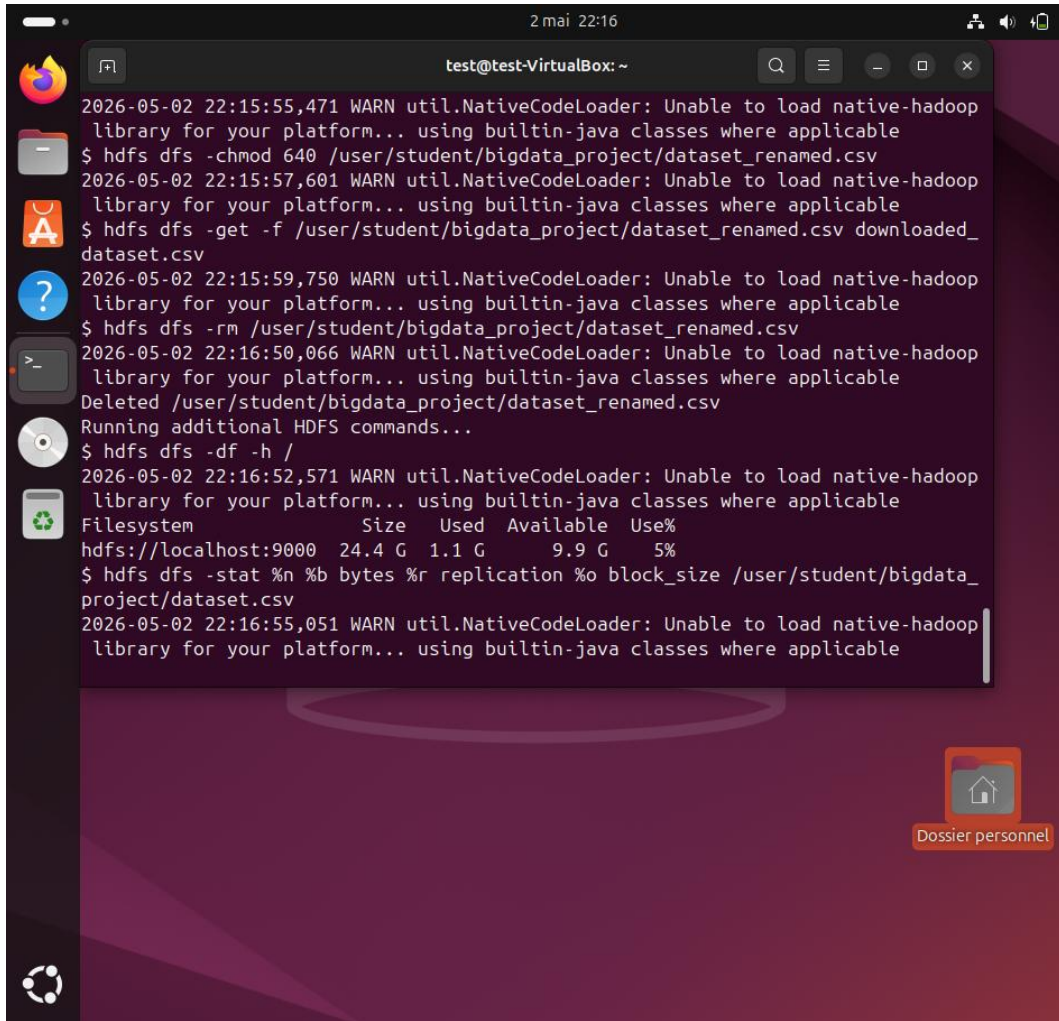
Additional HDFS Commands

This section demonstrates additional HDFS utilities for metadata inspection, integrity verification, file creation, and content manipulation.

Command	Purpose / Expected Result
<code>hdfs dfs -df -h /</code>	Displays HDFS filesystem capacity, used and available space
<code>hdfs dfs -stat "%n %b bytes %r replication %o block_size" ...</code>	Displays selected metadata for the uploaded file
<code>hdfs dfs -checksum .../dataset.csv</code>	Computes the HDFS checksum for data integrity verification
<code>hdfs dfs -test -e .../dataset.csv && echo exists</code>	Tests whether the file exists and prints confirmation
<code>hdfs dfs -touchz .../empty_marker.txt</code>	Creates an empty file in HDFS
<code>hdfs dfs -appendToFile ...</code>	Appends local content to an HDFS file
<code>hdfs dfs -getmerge -nl ...</code>	Merges HDFS files into one local output file
<code>hdfs dfs -find /user/student/bigdata_project -name '*.csv'</code>	Finds all CSV files in the project directory

Script Execution and df/stat Commands

The project script automates the execution of all HDFS commands. The screenshot below shows the `chmod` operation, file download, deletion confirmation, and the start of additional commands including `df` (showing 24.4G total capacity with 1.1G used) and the `stat` command for metadata inspection.



```

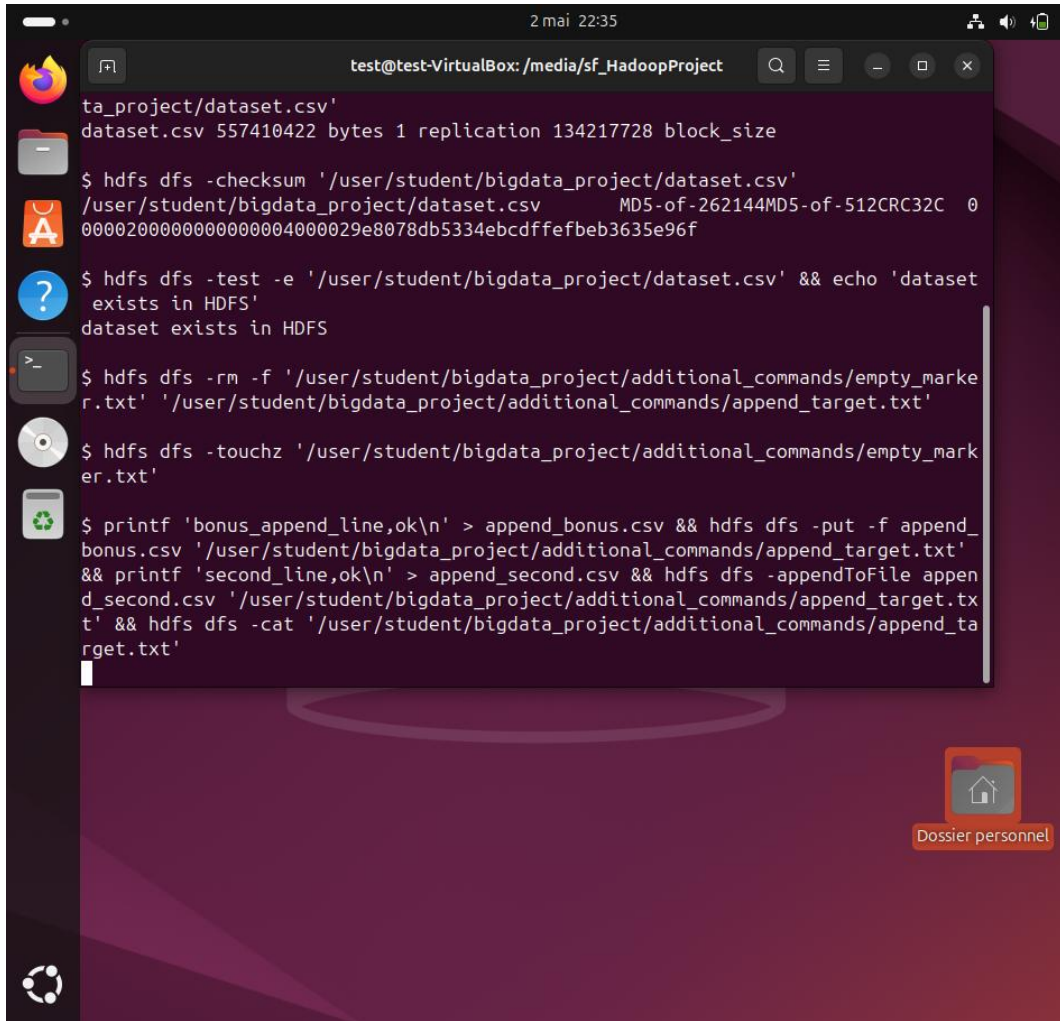
2026-05-02 22:15:55,471 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
$ hdfs dfs -chmod 640 /user/student/bigdata_project/dataset_renamed.csv
2026-05-02 22:15:57,601 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
$ hdfs dfs -get -f /user/student/bigdata_project/dataset_renamed.csv downloaded_
dataset.csv
2026-05-02 22:15:59,750 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
$ hdfs dfs -rm /user/student/bigdata_project/dataset_renamed.csv
2026-05-02 22:16:50,066 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Deleted /user/student/bigdata_project/dataset_renamed.csv
Running additional HDFS commands...
$ hdfs dfs -df -h /
2026-05-02 22:16:52,571 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Filesystem              Size      Used Available  Use%
hdfs://localhost:9000  24.4 G   1.1 G    9.9 G     5%
$ hdfs dfs -stat %n %b bytes %r replication %o block_size /user/student/bigdata_
project/dataset.csv
2026-05-02 22:16:55,051 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable

```

Figure 9: Project script execution showing `chmod`, `get`, `rm`, and `df` commands

Stat, Checksum, Test, and Append Commands

This screenshot demonstrates the `stat` command output showing `dataset.csv` metadata (557,410,422 bytes, 1 replication, 128 MB block size), the MD5-of-CRC32C checksum for integrity verification, the existence test confirming the file is present, and the `touchz/appendToFile` workflow for content manipulation.



```

test@test-VirtualBox: /media/sf_HadoopProject
ta_project/dataset.csv'
dataset.csv 557410422 bytes 1 replication 134217728 block_size

$ hdfs dfs -checksum '/user/student/bigdata_project/dataset.csv'
/user/student/bigdata_project/dataset.csv      MD5-of-262144MD5-of-512CRC32C  0
0000200000000000000000000000000029e8078db5334ebcdfefeb3635e96f

$ hdfs dfs -test -e '/user/student/bigdata_project/dataset.csv' && echo 'dataset
exists in HDFS'
dataset exists in HDFS

$ hdfs dfs -rm -f '/user/student/bigdata_project/additional_commands/empty_marke
r.txt' '/user/student/bigdata_project/additional_commands/append_target.txt'

$ hdfs dfs -touchz '/user/student/bigdata_project/additional_commands/empty_marke
r.txt'

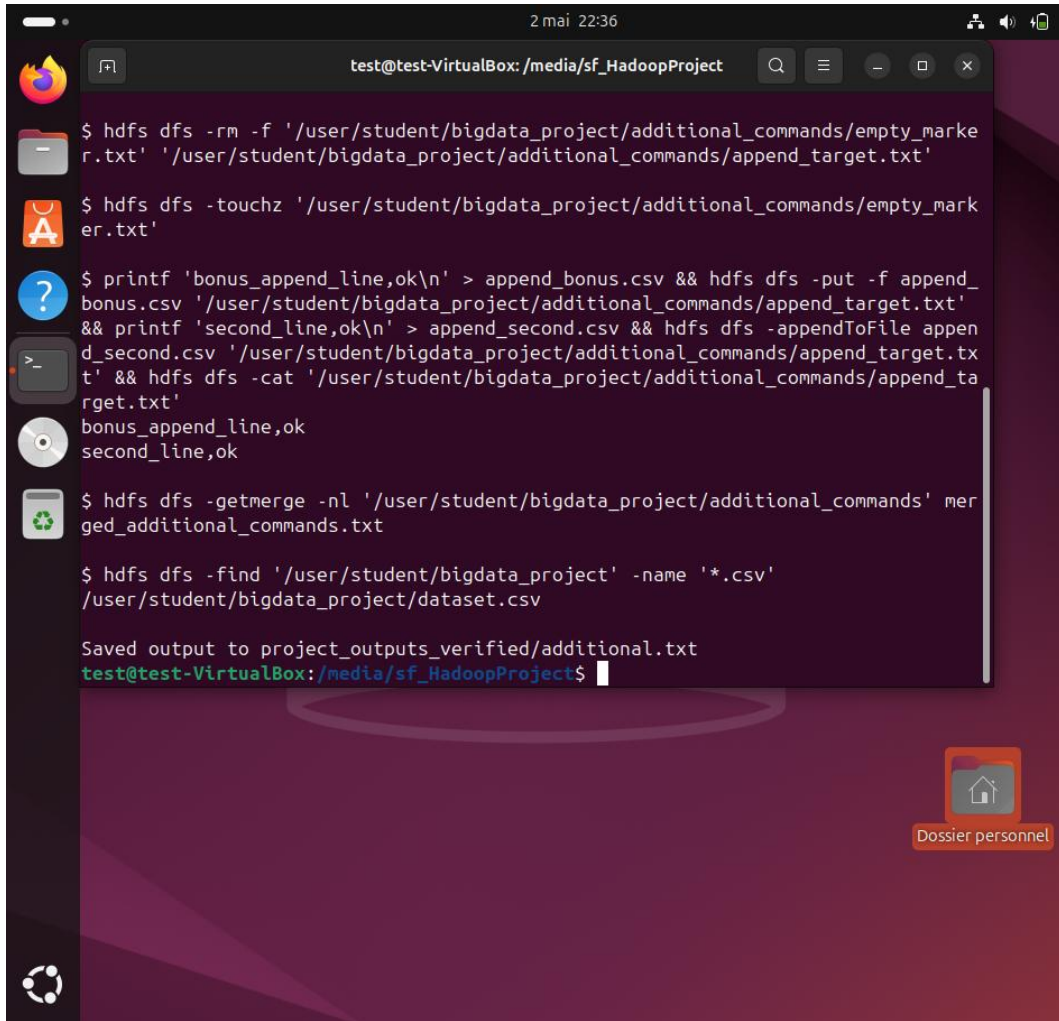
$ printf 'bonus_append line,ok\n' > append_bonus.csv && hdfs dfs -put -f append_
bonus.csv '/user/student/bigdata_project/additional_commands/append_target.txt'
&& printf 'second_line,ok\n' > append_second.csv && hdfs dfs -appendToFile appen
d_second.csv '/user/student/bigdata_project/additional_commands/append_target.tx
t' && hdfs dfs -cat '/user/student/bigdata_project/additional_commands/append_ta
rget.txt'

```

Figure 10: Additional commands - stat, checksum, test, touchz, and append operations

GetMerge and Find Commands

The final additional commands screenshot shows the getmerge operation which successfully merged HDFS files into a local file, and the find command which located all CSV files in the project directory.



```

test@test-VirtualBox: /media/sf_HadoopProject
$ hdfs dfs -rm -f '/user/student/bigdata_project/additional_commands/empty_marker.txt' '/user/student/bigdata_project/additional_commands/append_target.txt'
$ hdfs dfs -touchz '/user/student/bigdata_project/additional_commands/empty_marker.txt'
$ printf 'bonus_append_line,ok\n' > append_bonus.csv && hdfs dfs -put -f append_bonus.csv '/user/student/bigdata_project/additional_commands/append_target.txt' && printf 'second_line,ok\n' > append_second.csv && hdfs dfs -appendToFile append_second.csv '/user/student/bigdata_project/additional_commands/append_target.txt' && hdfs dfs -cat '/user/student/bigdata_project/additional_commands/append_target.txt'
bonus_append_line,ok
second_line,ok
$ hdfs dfs -getmerge -nl '/user/student/bigdata_project/additional_commands' merged_additional_commands.txt
$ hdfs dfs -find '/user/student/bigdata_project' -name '*.csv'
/user/student/bigdata_project/dataset.csv

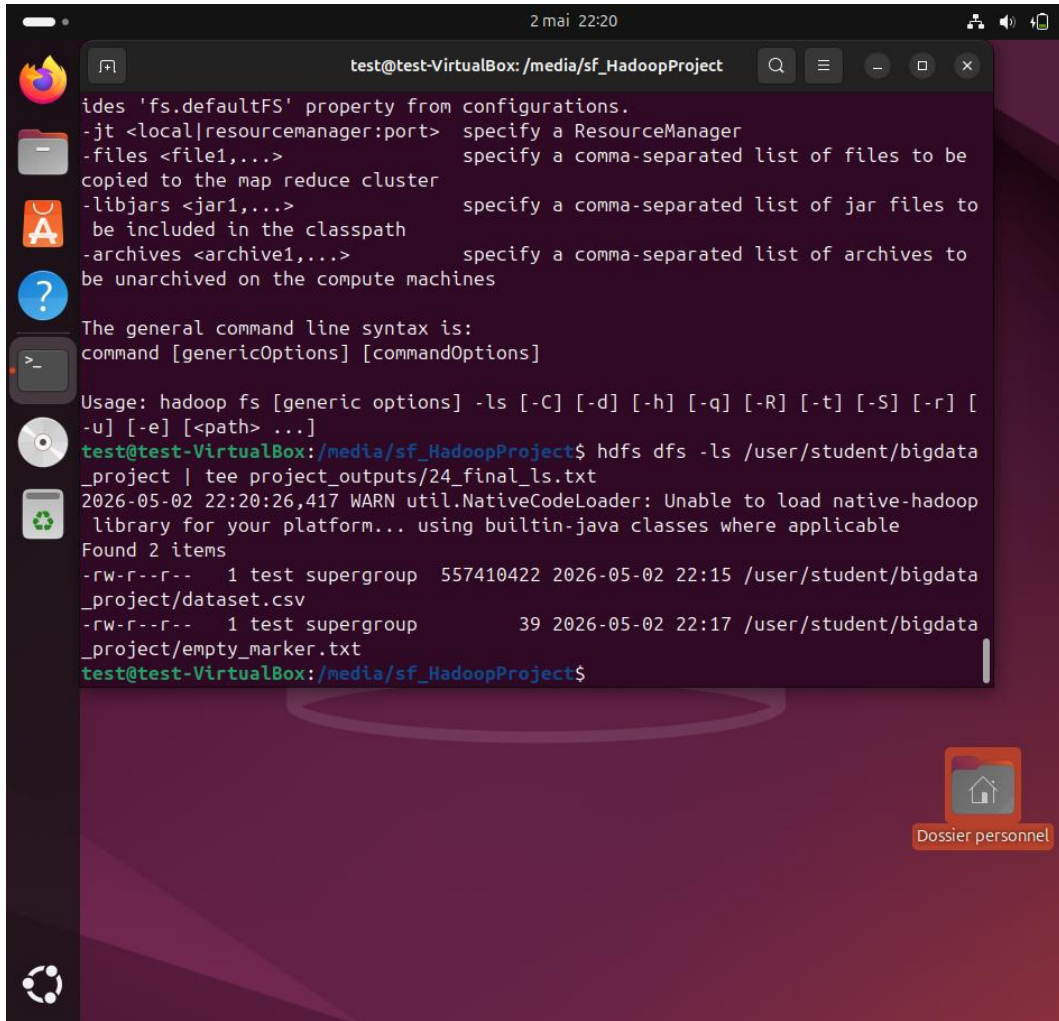
Saved output to project_outputs_verified/additional.txt
test@test-VirtualBox: /media/sf_HadoopProject$

```

Figure 11: Additional commands completed - getmerge output and find results for CSV files

Final HDFS Directory Listing

The final listing confirms the project directory contains the main dataset.csv file (557,410,422 bytes) and the empty_marker.txt file (39 bytes), demonstrating successful completion of all file management operations.



```

2 mai 22:20
test@test-VirtualBox: /media/sf_HadoopProject

ides 'fs.defaultFS' property from configurations.
-jt <local|resourcemanager:port> specify a ResourceManager
-files <file1,...> specify a comma-separated list of files to be
copied to the map reduce cluster
-libjars <jar1,...> specify a comma-separated list of jar files to
be included in the classpath
-archives <archive1,...> specify a comma-separated list of archives to
be unarchived on the compute machines

The general command line syntax is:
command [genericOptions] [commandOptions]

Usage: hadoop fs [generic options] -ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]

test@test-VirtualBox:/media/sf_HadoopProject$ hdfs dfs -ls /user/student/bigdata_project | tee project_outputs/24_final_ls.txt
2026-05-02 22:20:26,417 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 test supergroup 557410422 2026-05-02 22:15 /user/student/bigdata_project/dataset.csv
-rw-r--r-- 1 test supergroup 39 2026-05-02 22:17 /user/student/bigdata_project/empty_marker.txt
test@test-VirtualBox:/media/sf_HadoopProject$

```

Figure 12: Final HDFS directory listing showing dataset.csv and empty_marker.txt

Observed Results

The following table summarizes the key quantitative results observed during the project execution:

Command	Observed Result
<code>hdfs dfs -du -h /user/student/bigdata_project</code>	Dataset size in HDFS: 531.6 MB
<code>hdfs dfs -cat .../dataset.csv wc -l</code>	Line count: 2,950,001 (including CSV header)
<code>hdfs dfs -count /user/student/bigdata_project</code>	After bonus work: 8 directories, 7 files, 563,195,426 bytes total
<code>hdfs dfs -df -h /</code>	HDFS capacity: 24.4 GB total, 1.1 GB used, 9.9 GB available (5%)
<code>hdfs dfs -checksum .../dataset.csv</code>	MD5-of-CRC32C checksum: confirms file integrity in HDFS

Bonus Dataset Size Evidence

The screenshot below provides direct evidence of the large dataset size (531.6 MB), the line count (2,950,001), and the final directory count after all bonus tasks were completed.

```

test@test-VirtualBox: /media/sf_HadoopProject
BONUS LARGE DATASET EVIDENCE
2026-05-02 22:39:02,303 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
531.6 M 531.6 M /user/student/bigdata_project/dataset.csv
2026-05-02 22:39:07,059 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2950001
2026-05-02 22:39:18,593 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
8 7 563195426 /user/student/bigdata_project
test@test-VirtualBox: /media/sf_HadoopProject$

```

Figure 13: Bonus evidence - large dataset size (531.6 MB) and final directory statistics

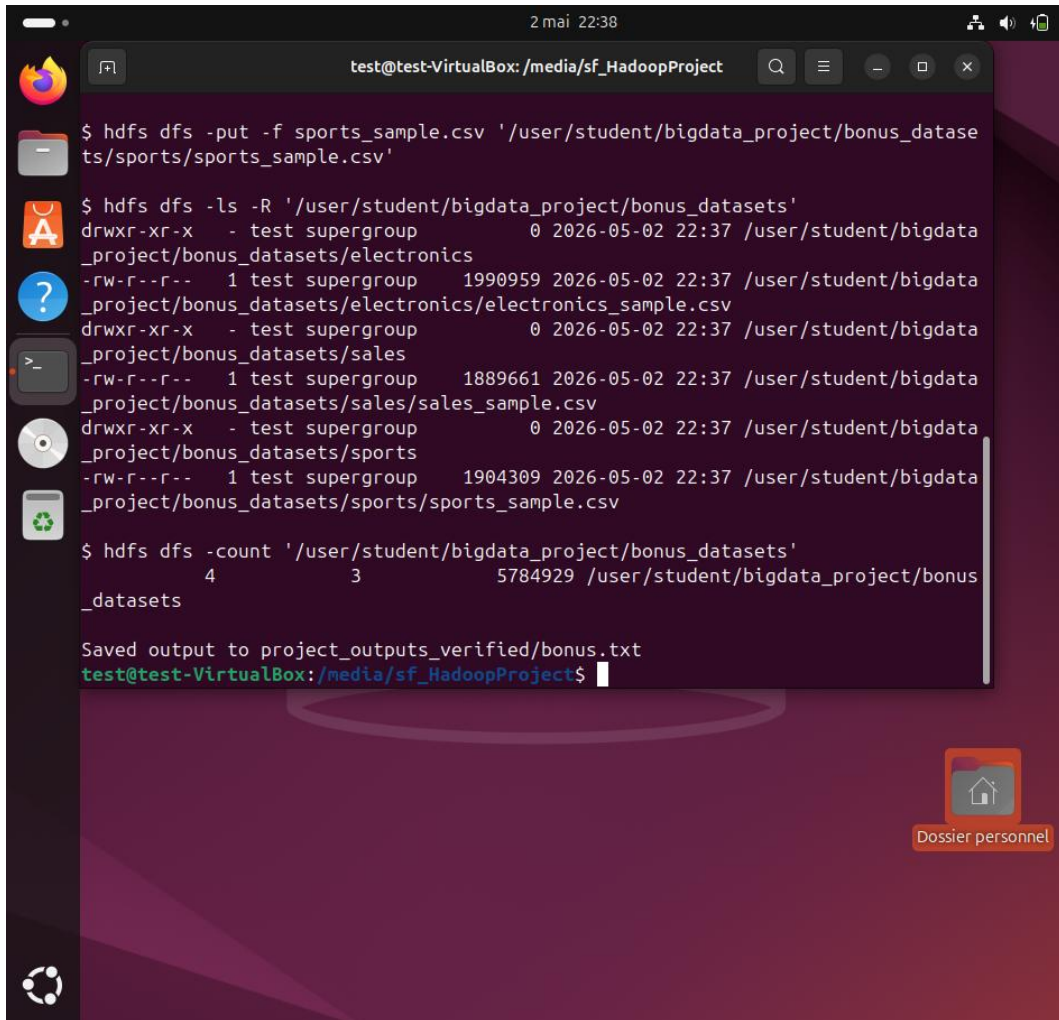
Bonus Tasks Completed

The project includes all three bonus options: a dataset larger than 100 MB, more than five additional HDFS commands, and directory management with multiple datasets.

Command	Purpose / Expected Result
Large dataset bonus	Main CSV is 531.6 MB in HDFS with 2,950,001 lines including header
Multiple commands bonus	Executed: mkdir -p, df -h, stat, checksum, test -e, touchz, appendToFile, getmerge, find, ls, echo
Multiple datasets bonus	Created sales, electronics, and sports sample datasets under /user/student/bigdata_project/bonus_datasets

Multiple Dataset Directories

The screenshot shows the recursive listing of `bonus_datasets` directory containing three subdirectories (`electronics`, `sales`, `sports`), each with a sample CSV file. The count command confirms 4 directories and 3 files in the `bonus_datasets` folder.



```

$ hdfs dfs -put -f sports_sample.csv '/user/student/bigdata_project/bonus_datasets/sports/sports_sample.csv'

$ hdfs dfs -ls -R '/user/student/bigdata_project/bonus_datasets'
drwxr-xr-x  - test supergroup          0 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/electronics
-rw-r--r--  1 test supergroup 1990959 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/electronics/electronics_sample.csv
drwxr-xr-x  - test supergroup          0 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/sales
-rw-r--r--  1 test supergroup 1889661 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/sales/sales_sample.csv
drwxr-xr-x  - test supergroup          0 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/sports
-rw-r--r--  1 test supergroup 1904309 2026-05-02 22:37 /user/student/bigdata_project/bonus_datasets/sports/sports_sample.csv

$ hdfs dfs -count '/user/student/bigdata_project/bonus_datasets'
4          3          5784929 /user/student/bigdata_project/bonus_datasets

Saved output to project_outputs_verified/bonus.txt
test@test-VirtualBox: /media/sf_HadoopProject$

```

Figure 14: Bonus evidence - multiple dataset directories and sample CSV files in HDFS

Complete VM Run with Bonus Commands

The final screenshot shows the complete execution of the project script including all bonus commands. It demonstrates the creation of bonus dataset directories, uploading of sample files, and the find command locating all CSV files across the entire project.

```

7 mai 22:14
test@test-VirtualBox: ~/hadoopdata

tin-java classes where applicable
Found 2 items
-rw-r--r-- 1 test supergroup 159601415 2026-05-07 22:11 /user/student/bigdata_project/dataset.csv
-rw-r--r-- 1 test supergroup 39 2026-05-07 22:12 /user/student/bigdata_project/empty_marker.txt

Bonus commands
$ hdfs dfs -mkdir -p /user/student/bigdata_project/bonus_datasets/sales
2026-05-07 22:13:30,597 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -mkdir -p /user/student/bigdata_project/bonus_datasets/electronics
2026-05-07 22:13:32,038 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -mkdir -p /user/student/bigdata_project/bonus_datasets/sports
2026-05-07 22:13:33,473 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -put -f sales_sample.csv /user/student/bigdata_project/bonus_datasets/sales/
2026-05-07 22:13:34,910 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -put -f electronics_sample.csv /user/student/bigdata_project/bonus_datasets/electronics/
2026-05-07 22:13:36,921 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -put -f sports_sample.csv /user/student/bigdata_project/bonus_datasets/sports/
2026-05-07 22:13:40,183 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
tin-java classes where applicable
$ hdfs dfs -find /user/student/bigdata_project -name *.csv
2026-05-07 22:13:41,949 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
/user/student/bigdata_project/bonus_datasets/electronics/electronics_sample.csv
/user/student/bigdata_project/bonus_datasets/sales/sales_sample.csv
/user/student/bigdata_project/bonus_datasets/sports/sports_sample.csv
/user/student/bigdata_project/dataset.csv
Done. Command outputs are in project_outputs/.
test@test-VirtualBox:~/hadoopdata$

```

Figure 15: Complete VM run showing bonus dataset creation and all project commands executed

Problems Encountered

Several challenges were encountered during the project execution:

1. Because the dataset is larger than 500 MB, upload and copy commands take more time than small lab files. The VM must have sufficient disk space before running `hdfs dfs -put`, `hdfs dfs -cp`, and `hdfs dfs -get` operations.
2. If Hadoop services are stopped, HDFS must be restarted before running commands. The `start-dfs.sh` and `start-yarn.sh` scripts initialize all required services.
3. Some commands that pipe a large HDFS file into `head` can print a harmless broken-pipe style message after the requested rows are displayed, because `head` closes the stream early. This is expected behavior and does not affect the command results.
4. Native Hadoop library warnings ("Unable to load native-hadoop library") appear frequently but are cosmetic and do not affect functionality.

Conclusion

This project successfully demonstrates practical use of Hadoop HDFS for storing and managing a large structured dataset. The commands covered include creating directories, uploading files, inspecting data with `cat`/`head`/`tail`, counting records, checking storage usage with `du` and `df`, filtering records with `grep`, managing permissions with `chmod`, copying and renaming files with `cp` and `mv`,

downloading files with `get`, deleting files with `rm`, and using additional HDFS utilities such as `stat`, `checksum`, `test`, `touchz`, `appendToFile`, `getmerge`, and `find`.

The synthetic e-commerce transactions dataset of 531.6 MB and 2.95 million rows provided an excellent test case for big data operations. All bonus tasks were completed successfully, including managing multiple sample datasets and executing more than ten additional HDFS commands beyond the basic requirements.

